

Drone Challenge (UAV)

2026 Safari Rescue - HS Drone UAV

Conference City and State:

Georgia TSA State Leadership Conference (SLC) 2026
Athens, Georgia

School/Chapter:

South Forsyth High School
Cumming, Georgia - 30041

Team Members:

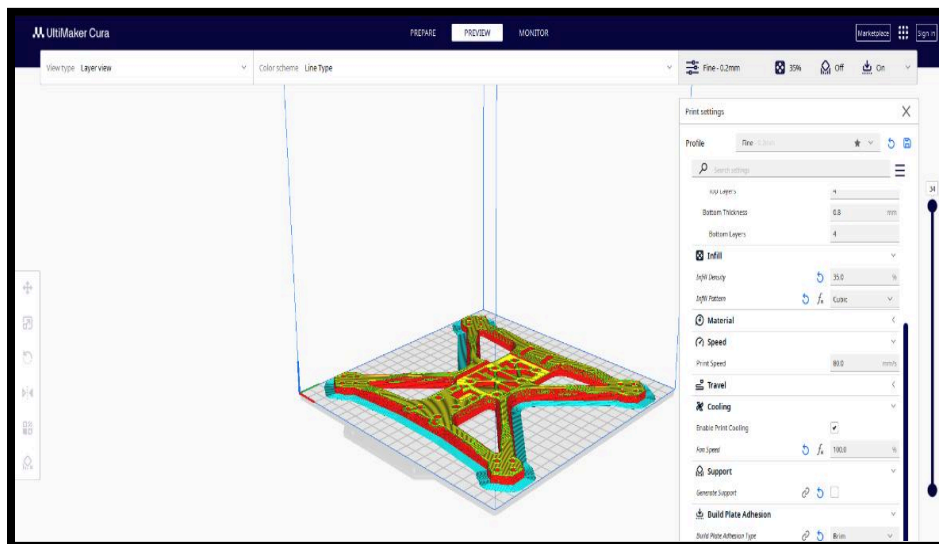
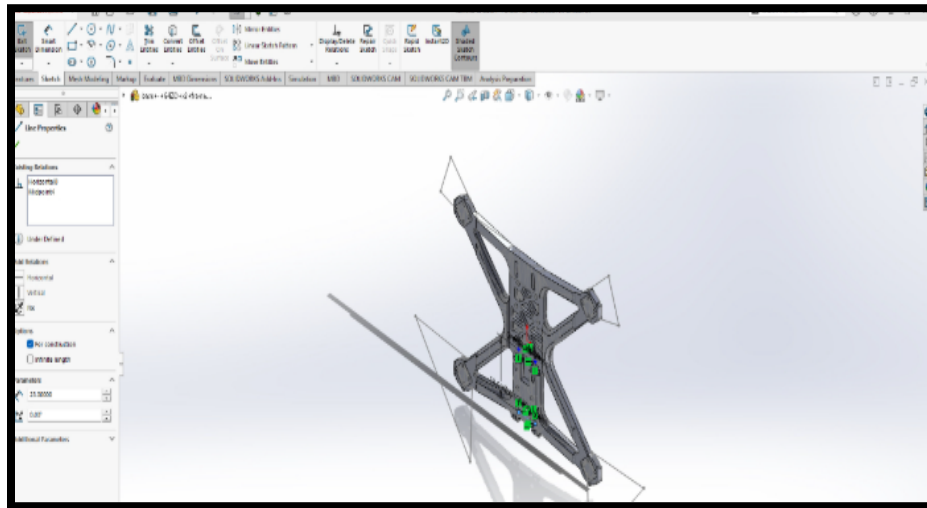
Benjamin Anthony
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Nathaniel Ramnath
Ivin George

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Photo Log

Photo 1 - Designing the Drone Frame on SolidWorks



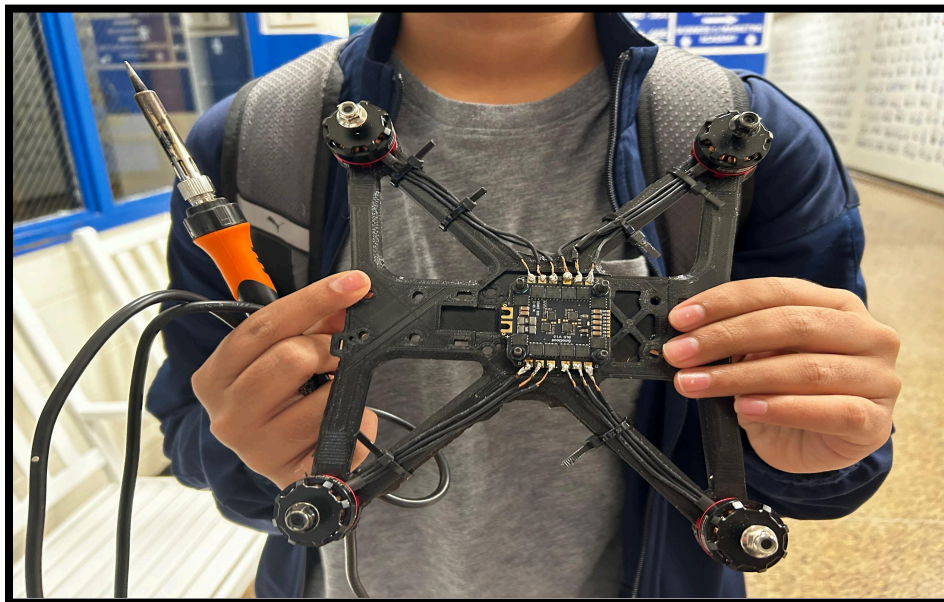
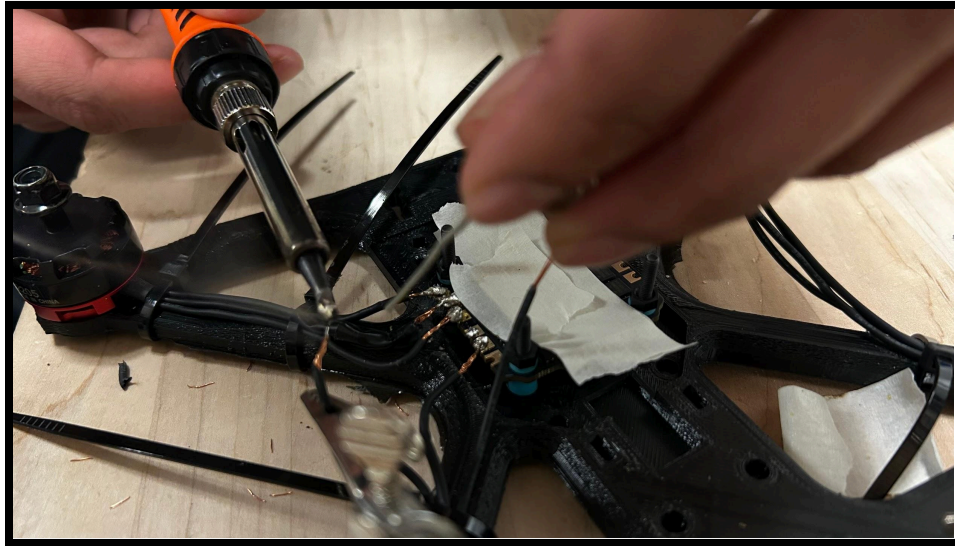
These images show the initial 3D modeling of the drone frame in SolidWorks. Designing the frame digitally allowed accurate planning before printing and ensured all components would fit properly.

Photo 2 - Attaching Motors to 3D-Printed Frame



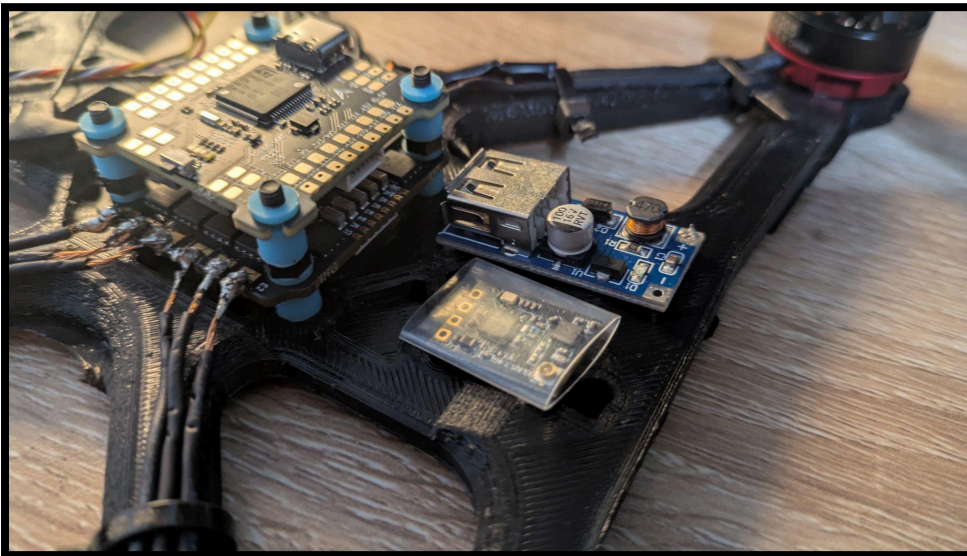
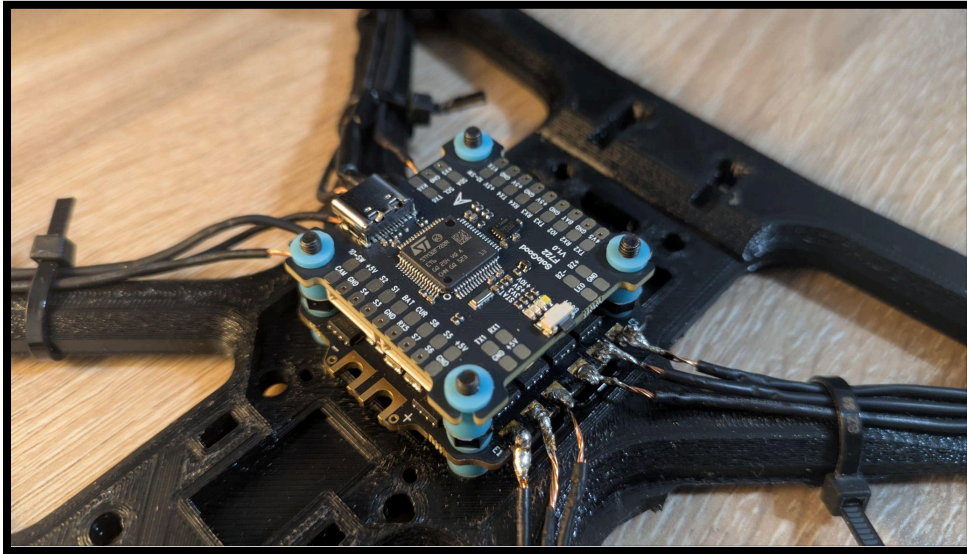
In this step, the motors were screwed onto the 3D-printed drone frame, ensuring each motor is securely mounted and ready for wiring.

Photo 3 - Soldering Motor Wires to the ESC



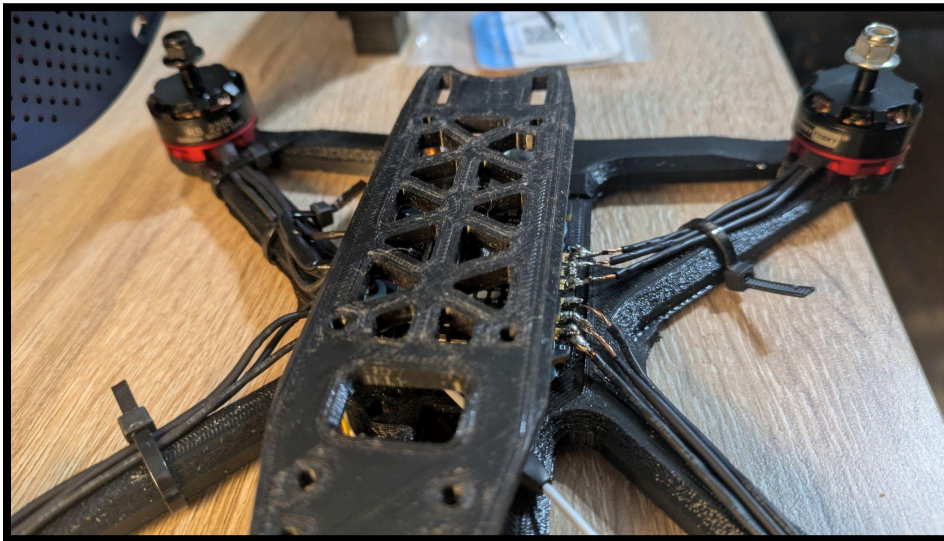
These photos show the soldering process where each motor's wires are connected to the Electronic Speed Controller (ESC).

Photo 4 - Installing Flight Controller and ELRS



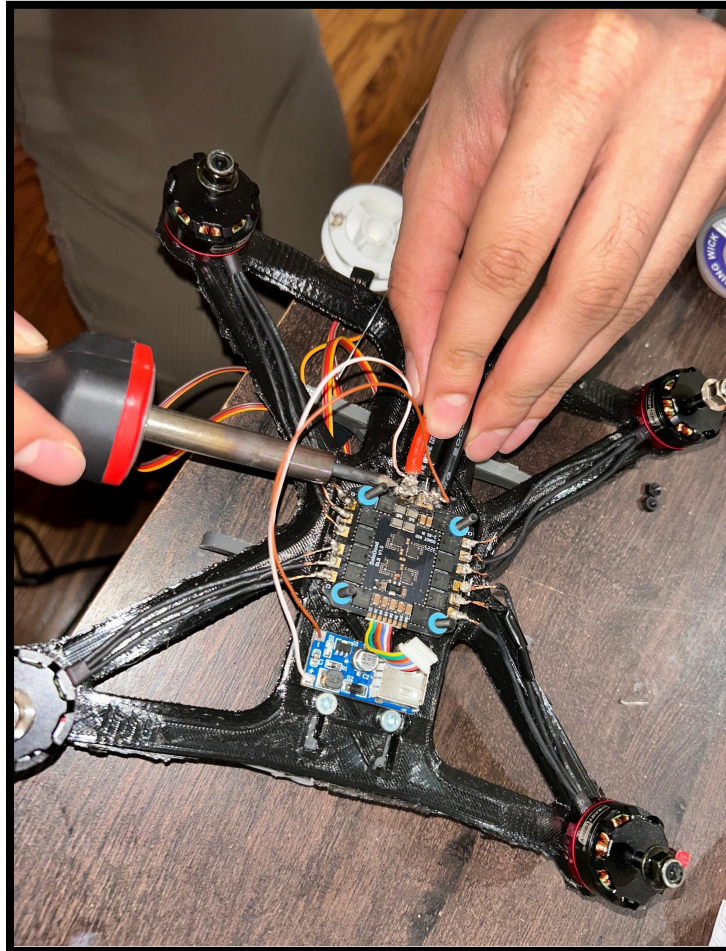
These photos show the installation of the flight controller and the ELRS receiver, completing the drone's control and communication system.

Photo 5 - Installing Top Frame



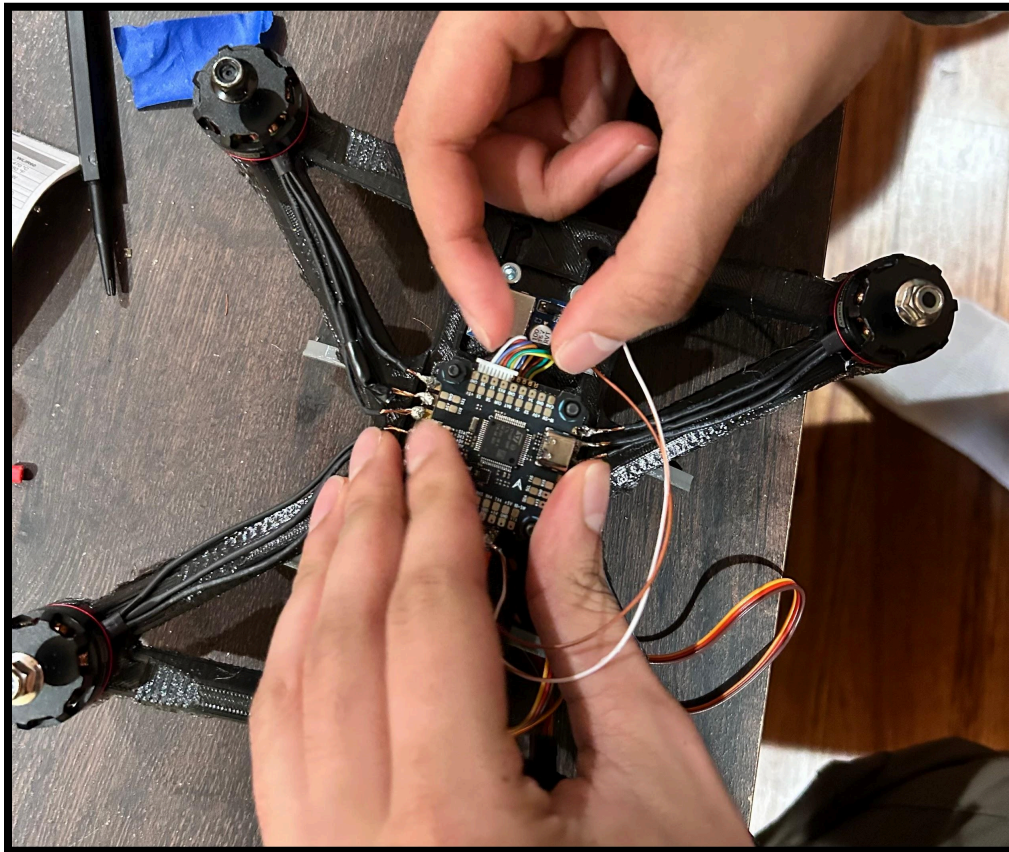
These photos show the installation of the 3D-printed top frame above the ESC, securing and protecting the components.

Photo 6 - Soldering battery wires to the ESC



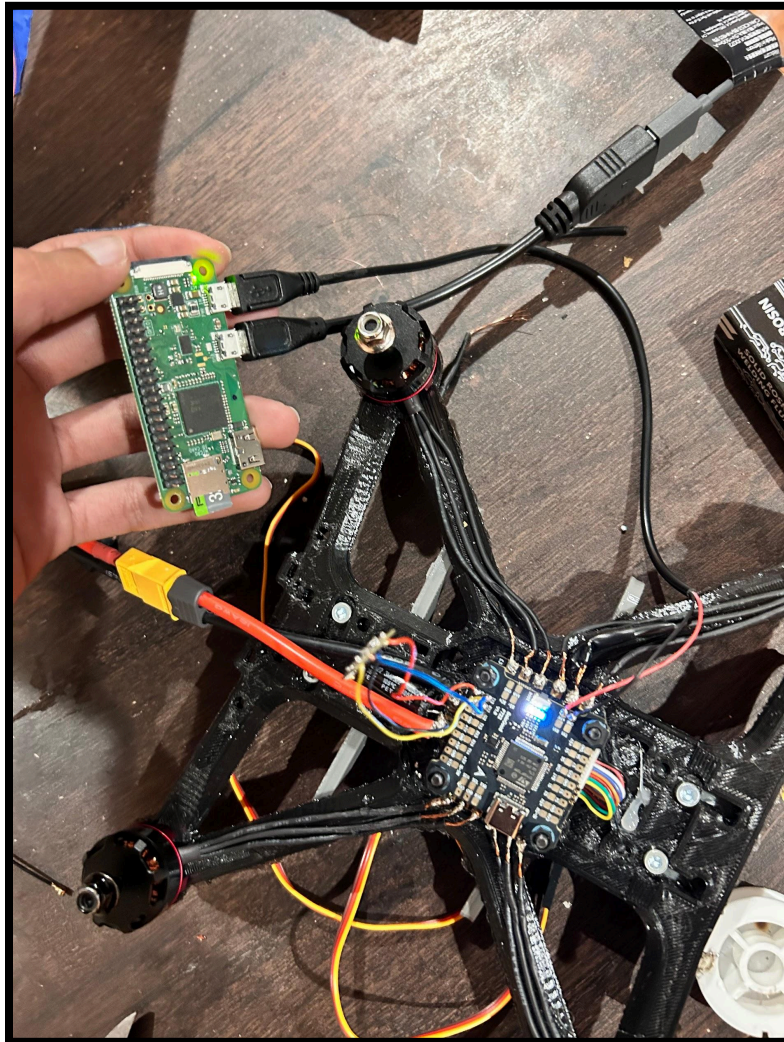
These photos show the soldering process where the drone's battery power leads (positive red and negative black wires) are connected to the ESC power input pads, ensuring current delivery to all four motors.

Photo 7 - Connecting Ribbon Cable to the Flight Controller



This photo shows the ribbon cable connection between the ESC and the flight controller for power and signal transmission.

Photo 8 - Connecting Ribbon Cable to the Flight Controller



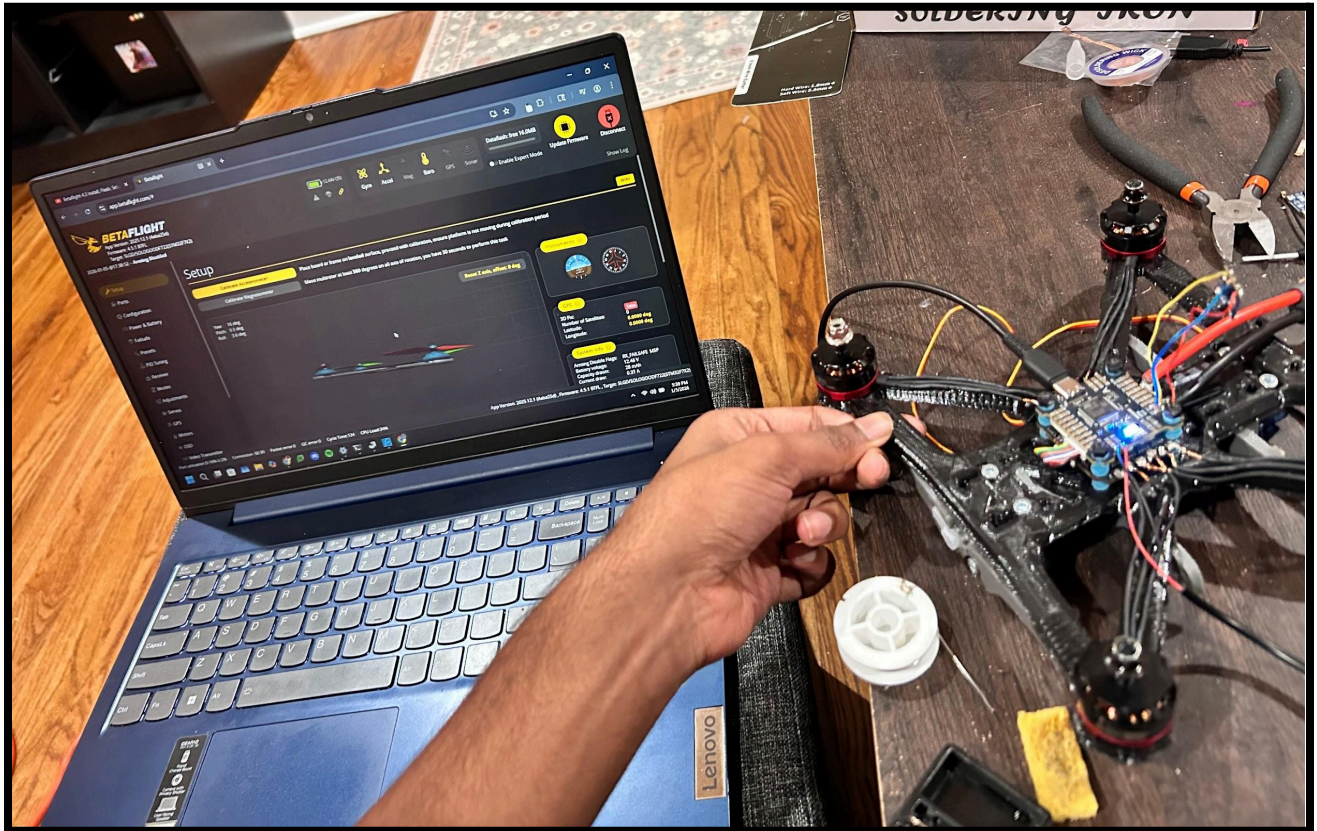
This photo shows the Raspberry Pi connected to the drone's flight controller to control the claw and webcam.

Photo 9 - Raspberry Pi and Webcam Mounted



This photo shows the Raspberry Pi and webcam securely mounted on the drone frame.

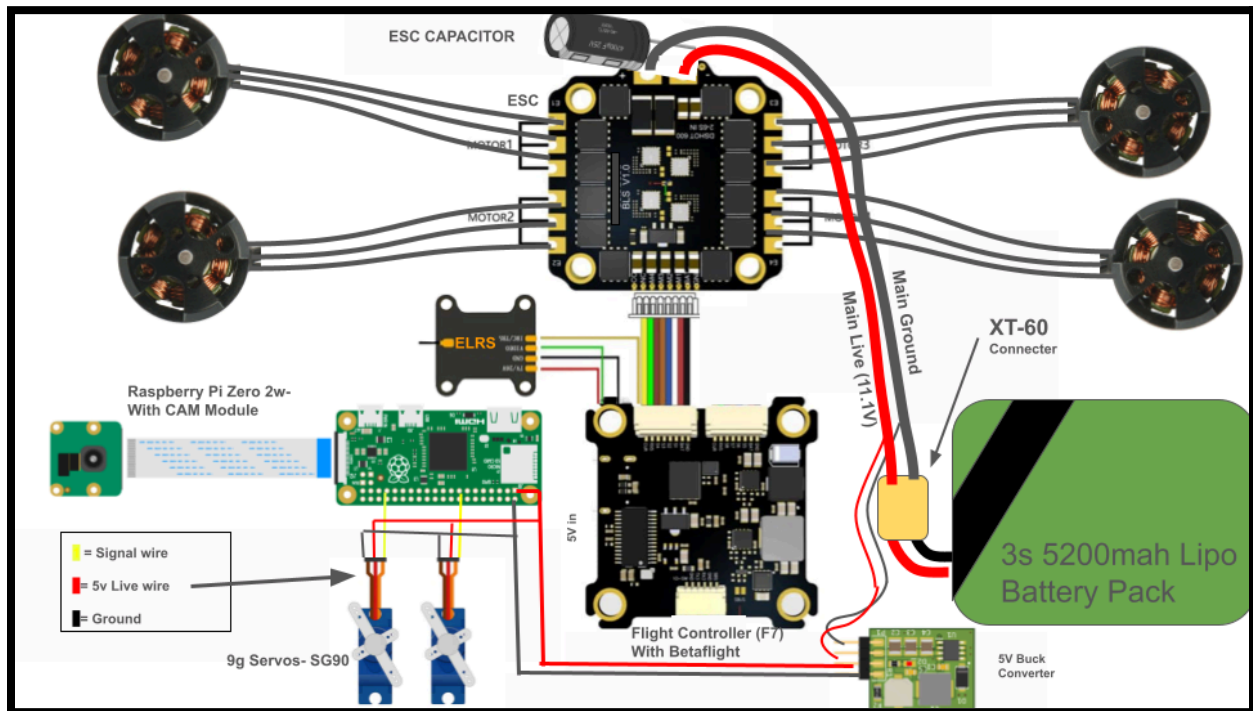
Photo 10 - Drone connected to Betaflight configurator



This photo shows the drone connected via USB to a laptop running Betaflight Configurator for flight control setup, receiver binding, controller and motor testing.

[Drone is currently under development. Photo log and other documentation will continue to be updated.]

Wiring Diagram



The information contained in this drawing is the sole property of Benjamin Antony, Taran Vijayakumar, Nathaniel Ramnath and Ivin George.

[Source of Flight Stack image: SoloGood FC Product Manual, used for illustrative purposes only]

Programming Software

Betaflight is an open-source flight control firmware that runs on the drone's flight controller microprocessor (in this build: STM32 F722 SoloGood stack) and acts as the central system coordinating all electronics. It continually processes pilot commands received via radio link (here using a BetaFPV ELRS 2.4 GHz receiver operating at 500Hz with the BetaFPV controller) while reading data from an Inertial Measurement Unit (IMU) containing a 3-axis gyroscope and 3-axis accelerometer.

For this Safari Rescue mission, we configured the drone to operate in Angle Mode, which provides automatic self-leveling for stability. Betaflight is good for our project because when the sticks are centered in Angle Mode, the outer loop commands zero tilt, causing the drone to actively self-level and maintain stable hover. This automatic stabilization is essential during our claw operations, as it eliminates the need for manual corrections and allows us to focus on collecting the target and claw positioning.

With Betaflight software, we will be able to tune our drone to our liking with many settings like PID, throttle response, and filters, which will allow the drone to fly optimally. The PID will be used for controlling how the vibrations from our brushless motors affect our drone in flight. This is extremely important to make sure our drone doesn't do anything that could cause harm to the components, which is exemplified by the fact that our frame is 3D printed.

Betaflight also allows us to configure the drone without manually making code for our specific motherboard, which will speed up the process of tuning and connecting the drone. Betaflight will also give us full control of everything connected to the motherboard, which includes servos, transmitter, and video broadcasting system.

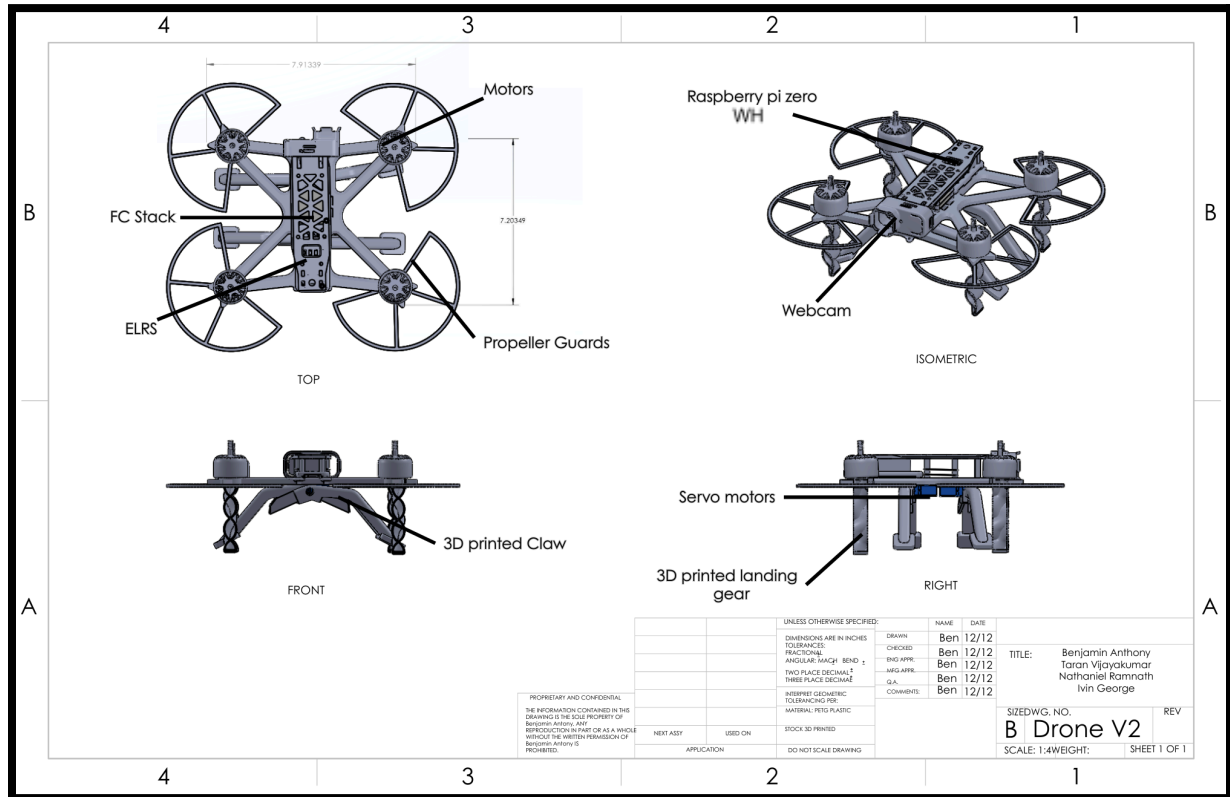
Our flight control loop also operates at high speeds: the ICM42688P gyroscope can sample up to 8 kHz, with PID calculations typically running at 4-8 kHz depending on processor load and enabled features. This is combined with the ELRS (ExpressLRS) radio control protocol, which works with the controller's pilot inputs sent to the ELRS receiver installed on the drone, with up to 500Hz update rate. Rapid input transmission from

ExpressLRS and high frequency sensor feedback from the gyroscope contribute to the drone's extremely low latency.

We will be using Betaflight in conjunction with a Raspberry Pi Zero WH (which has its own power rail) to operate the claw system and transmit the video signal to our laptop. The Pi will be running Raspberry Pi OS, which is a custom version of Linux. This will connect to the signal wires for our servos and our USB webcam via the USB OTG port included in the microcomputer. The Betaflight software will also have control of our BetaFPV ELRS system, which connects the drone to the remote control system. This will allow us to tune our telemetry signal as well, and get the most performance out of our ELRS system for our specific use case.

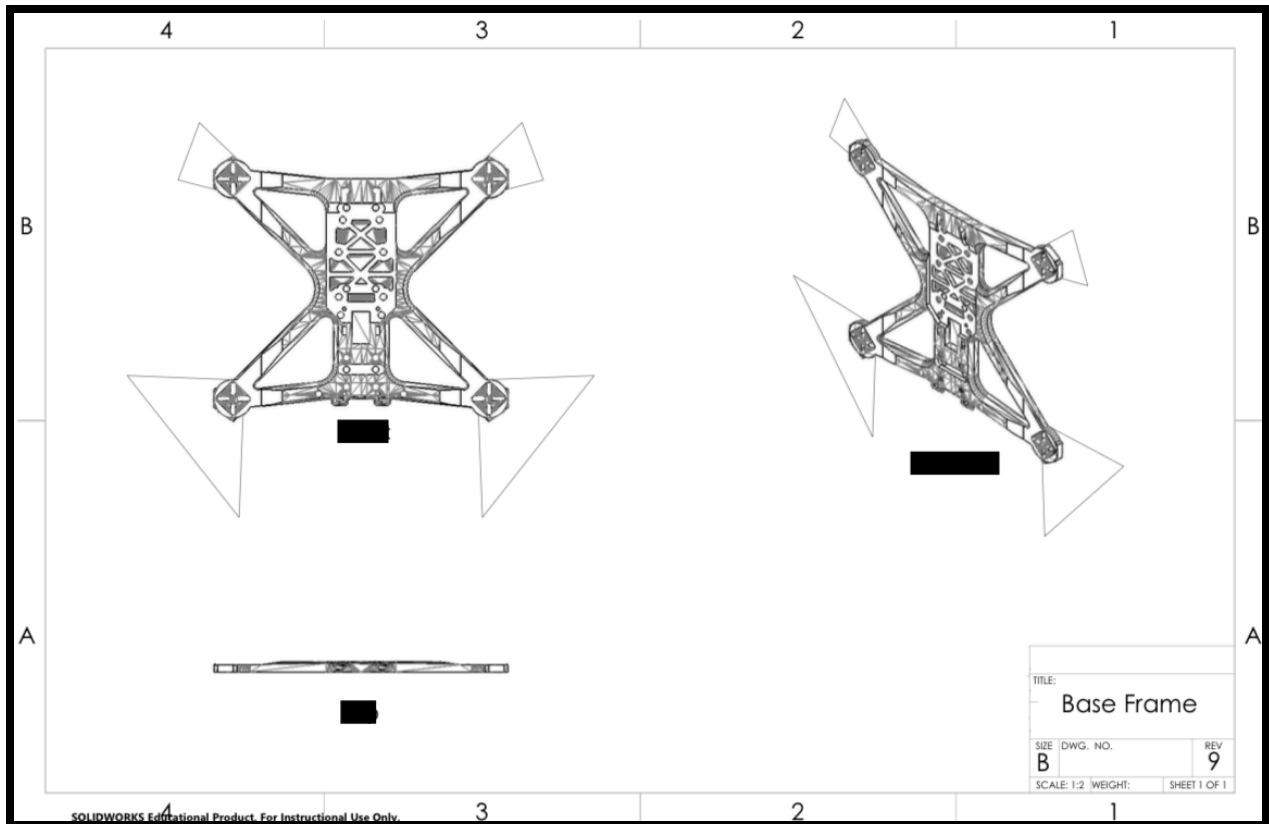
Essentially, Betaflight is the flight control firmware that manages and allows all the major parts of the drone to work together.

Engineered Drawings



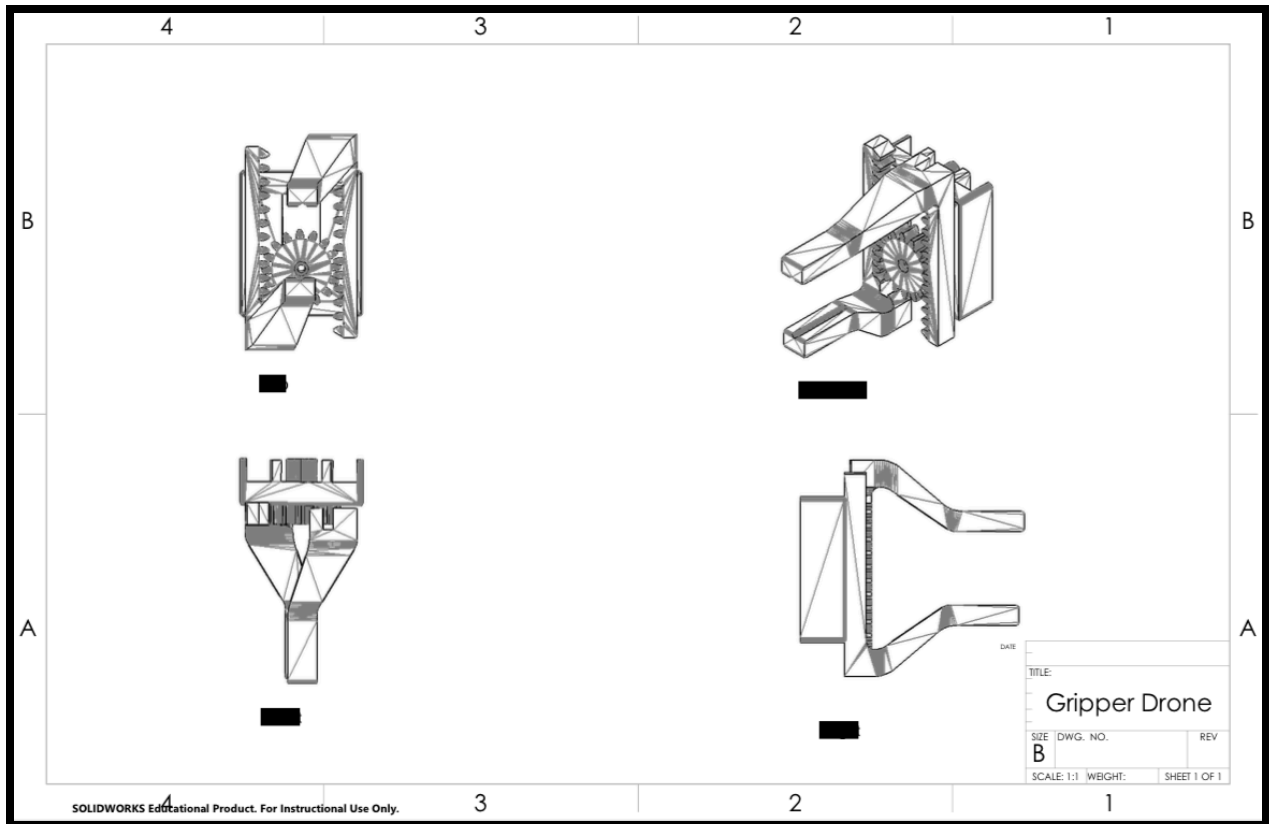
This engineered drawing shows the base frame, propeller and motor system, Raspberry Pi, claw, webcam, and all electronics. It shows the complete Safari Rescue drone configuration.

Base Frame



This engineered drawing shows the base frame with motor mounting holes and electronics mounting points. The design holds the motors, flight controller, ESC, and battery securely in place. It provides the main structure that connects all major components together for stable flight.

Gripper/Claw



This engineered drawing shows the claw mechanism and grip design. It holds and releases the caged and uncaged animals for the safari rescue mission.

The information contained in these drawings are the sole property of Benjamin Antony, Taran Vijayakumar, Nathaniel Ramnath and Ivin George. Also reference the links in the Resources page.

Bill of Materials

Parts	Description	Quantity
Anycubic Kobra 2	3D Printer	1
Sunlu PETG	Filament for 3D printer	4
Drone Frame	3D-printed drone frame - It supports and protects all drone components, with the lower frame mounting the ESC and the upper frame supporting the flight controller, Raspberry Pi, and claw system.	2
XT30/XT60 FPV Smoke Stopper	Short circuit protection - Protects stack and Raspberry Pi from catching on fire and protects electronic components from current surges.	1
Raspberry Pi Zero WH	For handling the video feed and servo motors	1
OVONIC D15 Dual LiPo Battery Charger	Charges the battery	1
3S LiPo 11.1V 5000mAh battery	Powers the drone and associated systems	2
Beta FPV controller	For controlling and piloting the drone	1
Beta FPV ELRS (0.7 g, 2.4 GHz, 5 V, 500Hz)	Sends signal from the drone to the controller	1
SoloGood F722 Stack	It acts like the motherboard of the drone, it determines motor speed and interprets signals	1

Readytosky RS2205 2300KV Brushless Motor CW/CCW 3-4S RC Motors	Brushless motor that spins propellers to generate lift and provide thrust for flight.	4
5.1inch, 3-Paddle Propellers	Creates lift to fly the drone	12
Webcam	Provides a camera feed and target identification	1
SG90 Micro Servos (from open-lab)	Powers the claw to pick up objects	3
5V Converter (from open-lab)	5V regulator to power Raspberry Pi and servos.	1
Hand drill	Used for creating mounting holes to attach motors and other components to drone frame	1
Claw	3D printed and used to pick up objects	1

Rules & Regulations

Federal Rules and Regulations:

In accordance with the Federal Aviation Administration (FAA) regulations, any unmanned aircraft weighing more than 250 grams must be registered with the FAA and must comply with RFID (Remote ID) requirements unless operated within a FAA-Recognized Identification Area (FRIA). Pilots must ensure that the aircraft's Remote ID broadcast is functional prior to takeoff, as it transmits the drone's location and registration information for accountability and airspace safety.

The maximum allowable altitude for all drones is 400 feet above ground level. The aircraft must be maintained within Visual Line of Sight of the pilot or a dedicated spotter at all times. FPV (First Person View) operations are allowed only when a visual observer maintains unaided visual contact with the drone. Prior to flight, pilots are required to check the FAA B4UFLY application for airspace restrictions, such as those found near airports, military bases, or other controlled airspace. Drone flight is prohibited over national parks, and large gatherings of people, unless specific authorization is granted.

Operations near critical infrastructure such as power plants, hospitals, and transmission lines are discouraged unless explicitly approved. All pilots are required to complete the FAA TRUST exam and have the required certification. The drone must not interfere with emergency response, law enforcement, or firefighting activities. The drone may not drop or release objects in a manner that creates a hazard to people or property. Flights should be conducted during daylight or civil twilight hours in accordance with FAA safety guidelines.

Local/Regional Rules and Regulations:

All operations comply with applicable Georgia state laws regarding unmanned aircraft systems and privacy. Operators are restricted from recording or photographing in areas where individuals have a reasonable expectation of privacy. Flight over state parks, public gatherings, correctional facilities, and using a drone to nuisance or annoy someone is strictly prohibited. Flights over private property without the property owner's consent are prohibited. Obtaining prior permission from local authorities or park services is required for any educational or research-based drone missions in these areas. Giving way to manned aircraft is also required as well as requiring authorization to fly near the local Athens-Ben Epps Airport (AHN). Operators should file a LAANC (Low Altitude Authorization and Notification Capability) request when flying within controlled airspace near airports to ensure compliance with FAA airspace management.

Resources & Sources

3D models used:

Claw arm:

<https://www.printables.com/model/1460464-replacement-arm-claw-for-pik-stik-grabber-tool/files>

Landing Gear:

<https://www.printables.com/model/67370-drone-landing-gear-sjrc-f11-pro>

FPV drone propeller guard:

<https://www.printables.com/model/1302713-5-fpv-drone-propeller-guard>

Raspberry Pi Flight Case:

<https://www.printables.com/model/703964-raspberry-pi-zero-2w-enclosure>

Servo Mockup:

<https://makerworld.com/en/models/1140652-sg90-micro-servo-9g-with-step#profileId-1142959>

Webcam Mockup:

<https://makerworld.com/en/models/1748017-runcam-thumb-2-fpv-camera-mount#profileId-1858095>

Drone Frame Files (ours is modified):

<https://makerworld.com/en/models/1336260-5-inch-fpv-drone>

Case for 5V Buck Converter:

<https://www.printables.com/model/326052-buck-converter-case>

Mockup of ESC and FC Board:

<https://www.printables.com/model/860808-mark4-drone-frame-stack-isolator>

Mockup For FPV drone motors:

<https://www.printables.com/model/665249-iflight-xing-2814-bldc-drone-motor>

Videos used:

BetaFPV binding: <https://youtu.be/3rdvve-Ecuc?si=N1mL7NIPLary5r7v>

Soldering tutorial: <https://www.youtube.com/watch?v=l5hilbGMowk>

Work Log

Date	Time spent	Team Members Present	Task/Activity	Materials Used	Progress	Next Steps
10/7/2025	1.5 hrs	Ben, Taran	Brainstormed potential designs for drone challenge	Laptops, Internet research	Figured out a clear roadmap and that we needed motors that can hold 1 kilogram of weight each	Find out parts online for budget of \$300 and more research for drone
10/21/2025 - 10/22/2025	3 hrs	Ben, Taran	Conducted research on drone designs	Laptops, and Research notes	Compiled data on materials and links to purchase design	Finalize list and then order parts in advance
10/25/2025 - 10/26/2025	5 hrs	Ben, Taran	Finalize compiled list of materials	Laptops and Research notes	Ordered list of needed materials for drone design	Start concept sketches and rough models.
10/27/2025 - 10/28/2025	4 hrs	Ben, Taran, Nathaniel, Ivin	Created concept sketches.	Solidworks 2024	Agreed on 3D design on Solidworks and finished drone sketch, and parts report	Begin Plan for drone and finalize checkpoint 0 submission
11/30/2025	4.5 hrs	Ben, Taran, Nathaniel, Ivin	Mounted motors to 3D-printed drone frame	Drill, Screws, Motors, and Drone frame	Mounted all 4 motors and began soldering	Solder motor wires to the ESC

12/1/2025 - 12/2/2025	3 hrs	Ben, Taran, Nathaniel, Ivin	Solder motor wires to the ESC and work on portfolio	Soldering iron, Solder, Wire strippers, and Helping hands with clamps	Finished soldering motor wires and worked on portfolio	Finish documentation portfolio
12/7/2025	2.5 hrs	Taran	Work on portfolio	Laptop, Research Notes, and Pictures	Finished first page,index, photo log, work log and worked on bill of materials	Finish documentation portfolio - engineered drawings and wired diagrams
12/8/2025	2 hr	Ben, Nathaniel, Taran	Find parts in open-lab and work on portfolio	Laptop, Scissors, and Battery to test converter, and open-lab	Found a 5V converter and wires for our drone and figured out how to wire the ELRS	Finish documentation portfolio - engineered drawings and wiring diagrams
12/10/2025	2.5 hrs	Ben, Taran	Work on engineered drawings	Laptop, Solidworks	Finished 3D-modeling of drone for the engineered drawings	Finish documentation portfolio - engineered drawings and wiring diagrams
12/12/2025	5.5 hrs	Ben, Taran, Ivin	Work on finishing the whole documentation portfolio	Laptop, Solidworks, Research Notes, and Pictures	Finished Documentation Portfolio and submitted for checkpoint 1	Finish building the drone and dropping mechanism.
1/4/2026	5 hrs	Ben, Taran, Nathaniel, Ivin	Work on setting up the drone for flight	Laptop, Soldering iron, Solder, Raspberry Pi Zero WH, SD card reader	Finished imaging and flashing the Raspberry Pi and soldered the battery.	Attach a webcam powered by the Raspberry PI to the drone and power motors.
1/5/2026	5 hrs	Ben, Taran, Nathaniel, Ivin	Finish mounting of parts onto the drone and start flying it	Laptop, Soldering iron, Solder, Raspberry Pi Zero WH, SD	Mounted the Battery, Webcam and Raspberry Pi Zero WH to	Bind the ELRS and the controller successfully

				card reader, BetaFPV Controller, and BetaFPV Configurator	the Drone frame	and begin flying the drone
1/6/2026	4 hr	Ben, Taran, Nathaniel, Ivin	Fix 3D printer and BetaFPV issues, add changes to documentation portfolio based on feedback, and begin flying the drone	Laptop, BetaFPV Controller, and BetaFPV Configurator	Added information to the drone flight regulations and work log sections of the documentation portfolio	Work on claw and fix BetaFPV issues
1/7/2026	3 hr	Ben, Taran, Nathaniel, Ivin	Fix 3D printer and motor issues	Laptop, BetaFPV Controller, and BetaFPV Configurator	Fixed 3D printer	Fix motor issues
1/9/2026	4 hr	Ben, Taran	Figure out how to make drone fly and finish changes to documentation portfolio	Laptop, BetaFPV Controller, and BetaFPV Configurator	Submitted updated documentation portfolio	Fix motor issues
1/11/2026	10 hr	Taran, Nathaniel, Ben, Ivin	Worked on claw and webcam, built the drone completely	Laptop, Soldering iron, Solder, Raspberry Pi Zero WH, SD card reader, BetaFPV Controller, and BetaFPV Configurator	Full drone completely assembled	Fix motor issues and raspberry pi
1/12/2026	7 hr	Taran, Nathaniel, Ben, Ivin	In house conference + resoldering	Drone	Identified motor issues and resoldered everything	Fix claw

Student Copyright Checklist

STUDENT: Answer question 1 below.

1) Does your solution to the competitive event integrate any type of music and/or sound?

☐ YES ☒ NO

- If NO, go to question 2.
- If YES, is the music and/or sound copyrighted?

☐ YES ☐ NO

- If YES, move to question 1A.
- If NO, move to question 1B.

1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation?

☐ YES ☐ NO

- If YES, move to question 2.
- If NO, ask for permission and, if permission is granted, include the permission in your documentation.

1B) Is the music/sound royalty-free, or did you create the music/sound yourself?

☐ YES

- If YES, cite the royalty-free music/sound or your original music/sound properly in your documentation.
-

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, _____ (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

2) Does your solution to the competitive event integrate any graphics and/or videos?

☒ YES ☐ NO

- If NO, go to question 3.
- If YES, is/are the graphics/videos copyrighted, registered, and/or trademarked?

☐ YES ☒ NO

- If YES, move to question 2A.
- If NO, move to question 2B.

2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for each graphic/video used?

☐ YES ☐ NO

- If YES, move to question 3.
- If NO, ask for permission and, if permission is granted, include the permission in your documentation.

2B) Is/are the graphics/videos royalty-free, or did you create your own graphics/videos?

☒ YES

- If YES, cite the royalty-free graphics/videos or your original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, _____ (chapter advisor), have checked my student's solution and confirm that any use of graphics/videos is done with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

3) Does your solution to the competitive event use another's thoughts or research?

☒ YES ☐ NO

- If NO, this is the end of the checklist.
- If YES, have you properly cited others' thoughts or research in your documentation?

☒ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, _____ (chapter advisor), have checked my student's solution and confirm that any use of the thoughts/research of others is done with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of others' thoughts/research.

Student Initials: TV / BA / NR / IG

Chapter Advisor Signature:
